

Computer Graphics –April 16th, 2025

Name: _____

Matriculation Number: _____

Please read the following instructions carefully:

- You have sixty (60) minutes to complete the exam.
- You can answer in German or in English.
- Check that the exam is complete (18 pages on 9 sheets) and readable!
- Please use a blue or black ballpen, not a red or green one. Do not use a pencil!
- Only text on the stapled sheets will be evaluated.
- If needed, we staple additional blank sheets.
- No other material except a calculator is allowed!

I have read and understood the instructions above.

Erlangen, April 16th, 2025

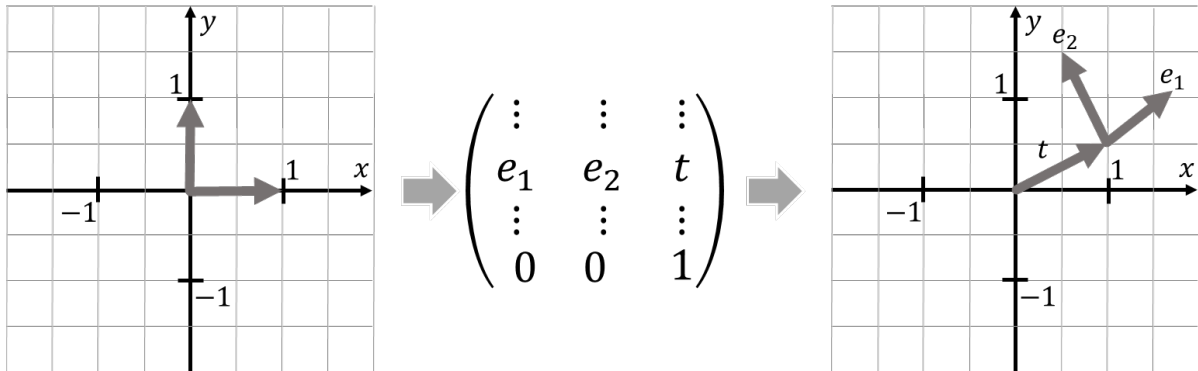
Signature

1	2	3	4	5	ALL
13	11	15	10	11	60

Grade:

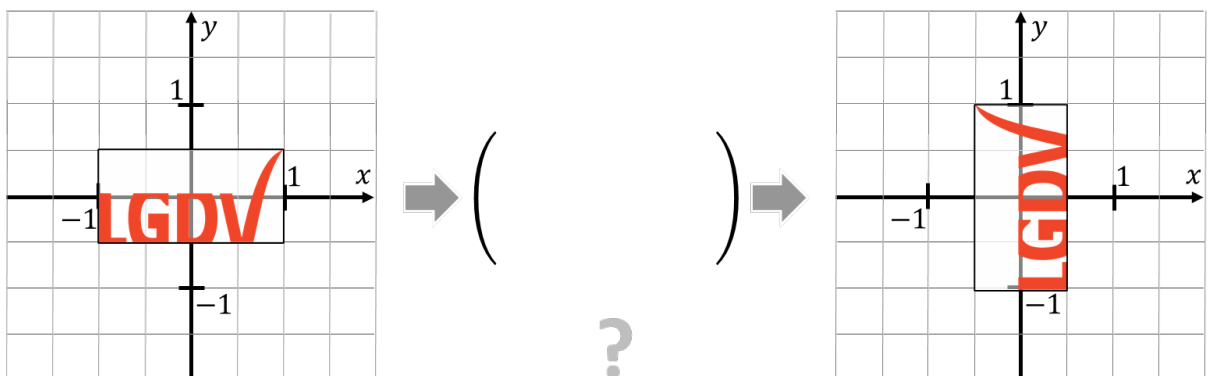
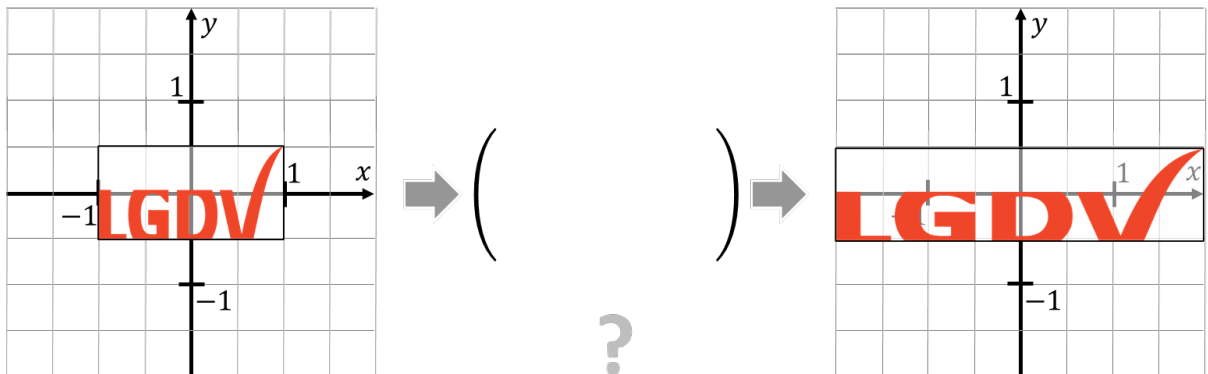
Assignment 1: Affine and Projective Transformations [13 points]

Affine mappings in **2D** can be expressed using **homogeneous coordinates** and 3×3 matrices with the following structure, where $(\vec{e}_1, \vec{e}_2)$ is the linear part (image of base vectors) and $\vec{t}$ is the translation:

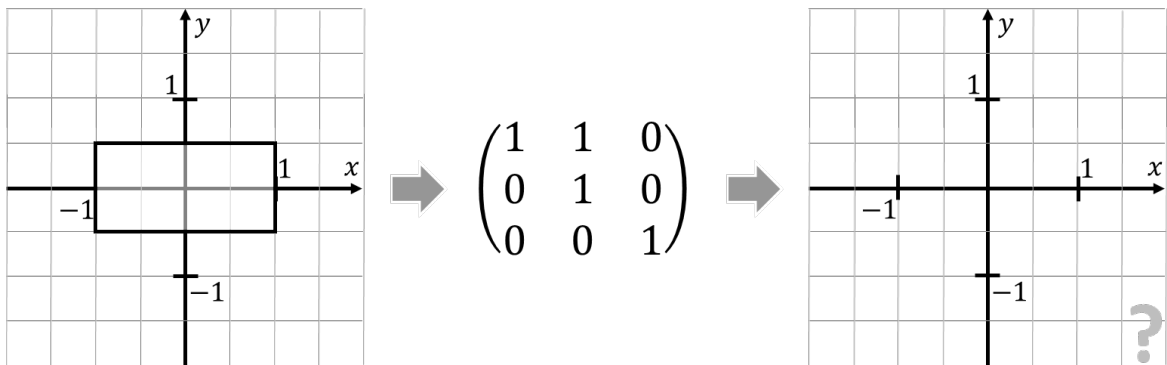
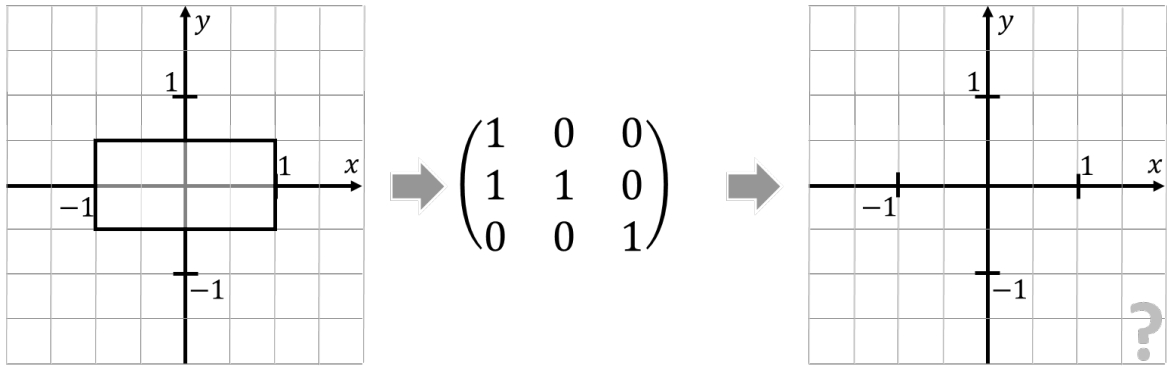


The following illustrations show a shape, which is transformed using an affine matrix.

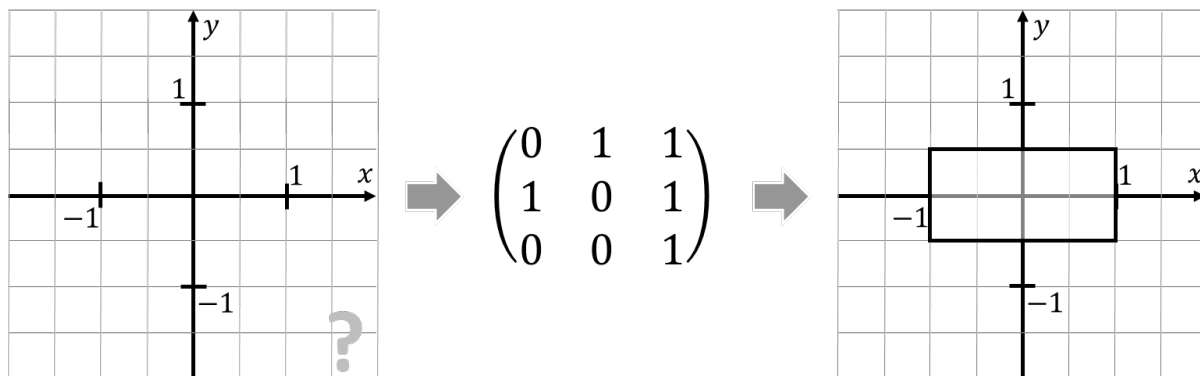
a) [2P] Fill in the matrix for the following mappings



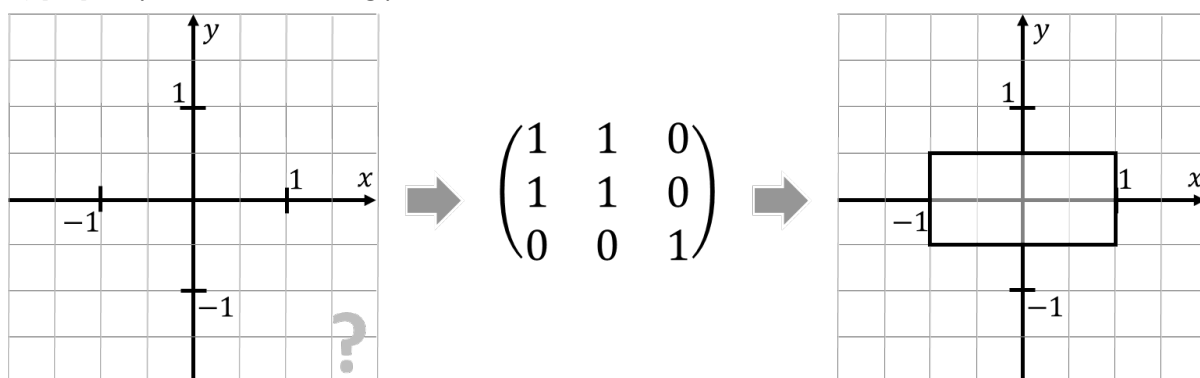
b) [2P] Sketch the result of the following mappings



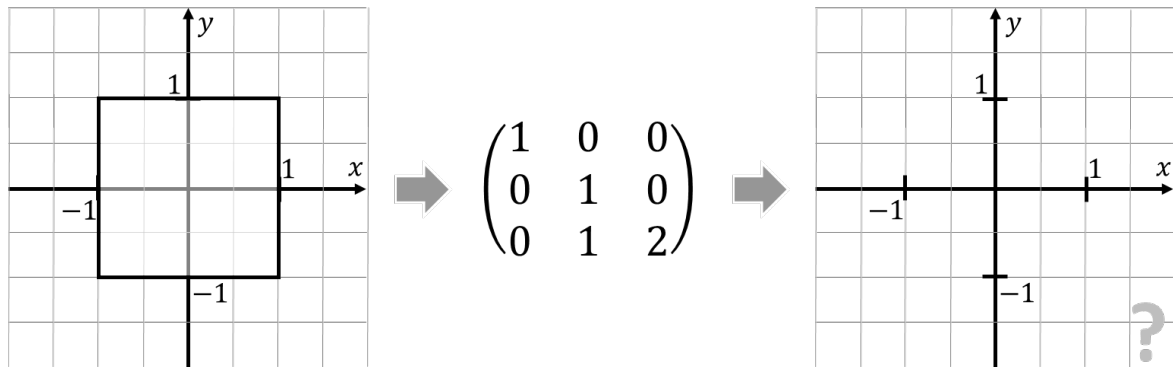
c) [2P] Sketch the source shape of the following mapping:



d) [1P] Why does the following problem have no solution?



e) [2P] The following matrix is **not affine**. Nevertheless, you can apply the mapping using homogeneous coordinates. Please sketch the result of the following projective mapping.



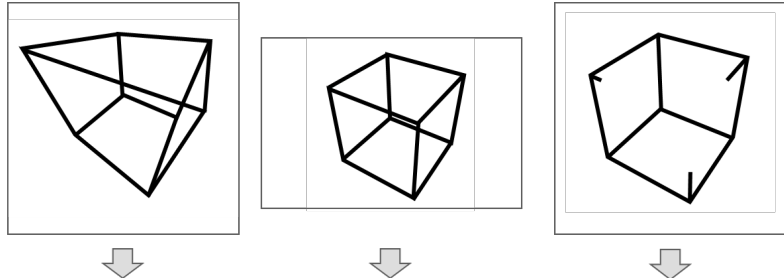
f) [1P] Describe the set of points (x, y) , which are mapped to infinity by the above matrix

g) [3P] Which of the following properties do affine and projective mappings have? Check the respective boxes or fill in yes/no.

	Affine	Projective
Parallel lines remain parallel		
Lines are mapped to lines		
Circles are mapped to circles		
Finite points are mapped to finite points		
Lengths are preserved		
Angles are preserved		

Assignment 2: Perspective [11 points]

a) **[3P]** Given are the following images of a cube with different camera parameters *fovy* (field of view in *y*-direction) and *aspect* (ratio of image width over height). Assign the correct parameter set to the images.



parameter	possible values	image 1	image 2	image 3
field of view	45° or 120°			
aspect ratio	1 or 1.5			
near	1 or 5			
far	100	100	100	100

b) **[2P]** Given the following perspective matrix:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.8 & 0 & 0 \\ 0 & 0 & -0.1 & -0.1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

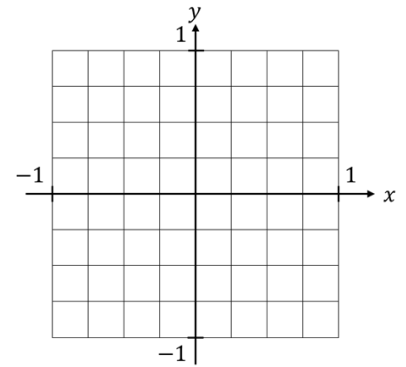
Is it a parallel perspective or a point perspective? Why?

b) **[4P]** Another camera has the following perspective matrix (which is different from a)

$$P = \begin{pmatrix} 0.5 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & 0 & -3 & -4 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

Map the rectangle with vertices $(1, -1, -1)$, $(1, -1, -2)$, $(1, 1, -2)$, $(1, 1, -1)$ using above matrix to image coordinates and sketch the 2D result.

$(1, -1, -1)$	→
$(1, -1, -2)$	→
$(1, 1, -2)$	→
$(1, 1, -1)$	→



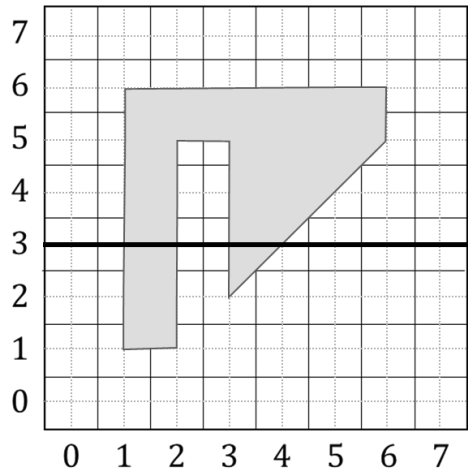
c) **[1P]** We now look at the *homogeneous* point $Q = (0, 0, 1, 0)$. Where is it positioned in the world?

d) **[1P]** What is the position of Q after the perspective mapping P ?

Assignment 3: Rasterization [15 points]

In this assignment, we look at the rasterization of polygons using the scanline algorithm.

a) [4P] Fill in the **edge table** for the following polygon (use integer coordinates).

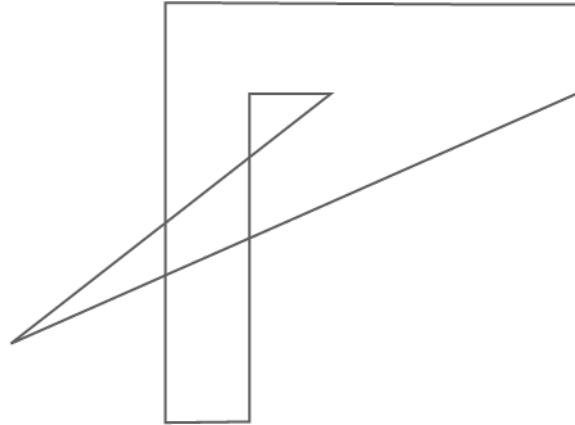
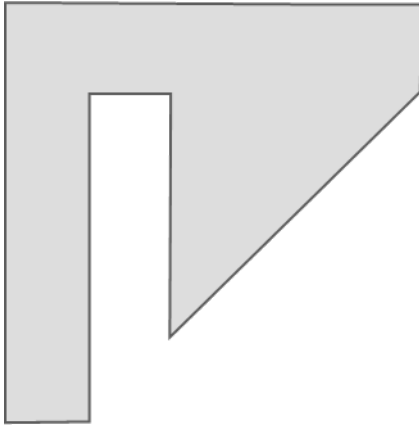


y_{lower}	x_{lower}	y_{upper}	$1/m$	$next$

b) [3P] Fill in the **active edge table** below for the scanline $y_{scan} = 3$

x_{inter}	y_{upper}	$1/m$	$next$

c) **[2P]** The shape above (left) is now modified by moving one vertex (right). Sketch in the right image, which regions will be filled.



d) [4P] Sketch an algorithm in pseudocode to rasterize a circle with center (m_x, m_y) and radius r , given a method **setPixel(x,y)** that sets a pixel at position (x, y) .

First, fill in the method **testInCircle(x,y,mx,my,r)** that tests, whether pixel (x, y) lies within the circle. Then, complete the code in **rasterizeCircle(mx,my,r)**.

```
void setPixel(x,y) { ... }
```

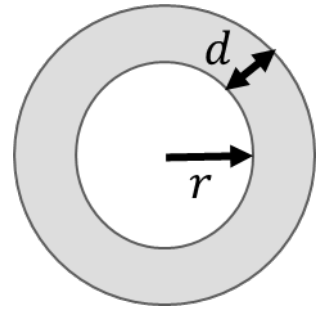
```
boolean testInCircle(x,y,mx,my,r)
```

```
{  
    return   
}
```

```
void rasterizeCircle(mx,my,r)
```

```
{  
    for (x =  )  
    {  
        for (y =  )  
        {  
            if (  )  
            {  
                setPixel(x,y)  
            }  
        }  
    }  
}
```

e) [2P] Write down pseudocode for a method **testInRing**(**x,y,mx,my,r,d**) that tests, whether pixel (x, y) is inside a ring with center (m_x, m_y) , inner radius r and thickness d .



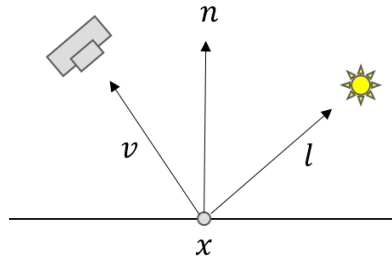
boolean testInRing (x,y,mx,my,r,d)

{

}

Assignment 4: (Blinn-)Phong Lighting & Shading [10 points]

Given is a surface point x with normal n , illuminated by a light source in direction l , and looked at from direction v , as shown in the image below. We now want to compute the lighting at x under incident light I_{in} , a diffuse coefficient of k_{diff} , a specular coefficient of k_{spec} , and a shiny exponent of n_{shiny} .



- a) **[1P]** Write down the formula to compute the diffuse lighting, using above defined variables. Use dot products to compute the cosine; you can assume that all direction vectors are normalized.

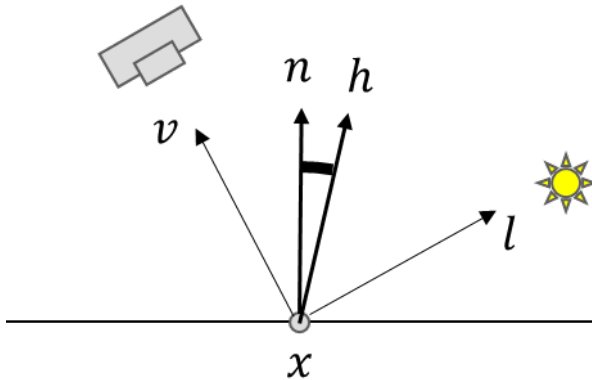
$$L_{diff} =$$

- b) **[2P]** Sketch the reflection vector r in the image above and write down the formula to compute the specular lighting (use Phong, not Blinn-Phong):

$$L_{spec} =$$

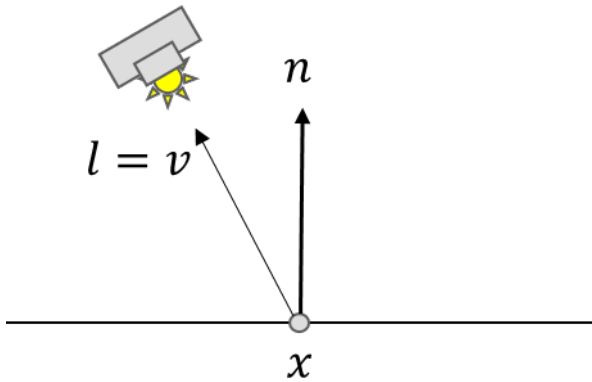
In the Blinn-Phong model, the “specular angle” is computed as the angle between the halfway vector and the normal:

$$L_{spec} = (n \circ h)^{n_{shiny}}$$



c) [2P] Write down the formula to compute the *normalized* halfway vector:

Now consider the case that the camera and light source are at the same position, i.e. $l = v$.



d) [2P] Compute diffuse (according to Phong) and specular lighting (according to Blinn-Phong) in this case, depending on n and v .

$$L_{diff}(n, v) =$$

$$L_{spec}(n, v) =$$

e) **[3P]** In a closed triangle mesh, the number of triangles T is roughly twice the number of vertices V : $T = 2V$. Assume we rasterize an object with T triangles, where each triangle covers P pixels. How often is lighting to be computed for Flat shading, Gouraud shading, and Phong shading?

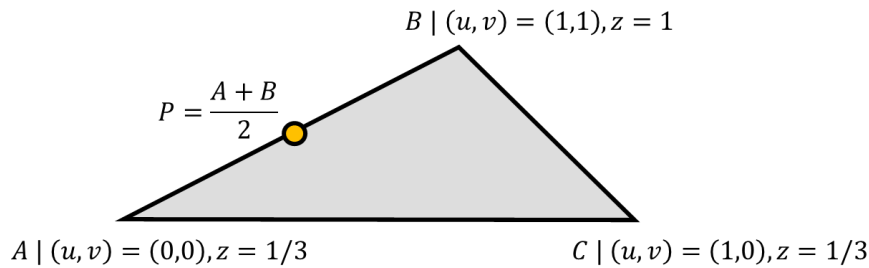
Use one of the following possible answers:

1 3 $\frac{T}{2}$ T $2T$ $4T$ P TP $3TP$

	Number of lighting evaluations
Flat Shading	
Gouraud Shading	
Phong Shading	

Assignment 5: Texture Mapping [11 points]

Given is a 2D triangle A, B, C in screen space (after perspective projection). Each vertex has texture coordinates (u, v) , plus a z -value from the original 3D triangle.



- a) **[1P]** P is the midpoint of A and B *after* perspective projection. What are the barycentric coordinates of P w.r.t. A, B, C in screen space?

$(\quad , \quad , \quad)$

- b) **[4P]** Perform a perspective correct texture coordinate interpolation for point P and write down the interpolated texture coordinates for P :

$$(u, v) = (\quad , \quad)$$

Next, let's assume the following image is a rendering, where the FAU-Logo is a square texture with 256x256 pixels. The texture is rendered with MIP-mapping. At the shown distance to the building of 60m, MIP-mapping selects exactly the resolution of 64x64 pixels.



c) [3P] Fill in the following table showing at which distance which texture resolution is selected by MIP-mapping:

Distance			60m			...
MIP-map resolution			64x64	32x32	16x16	...

d) [1P] At which distance range can we expect aliasing from magnification?

e) [2P] For the following perspective projections of the texture, check the boxes, where anisotropic MIP-mapping would be helpful.



